

Here are comments for 6d165343 on the file – *thesis.tex*:

1. p. 2 the add “and” after the centered equation for J_0 .
2. p. 2, Γ is the standard *Gamma-function* $\rightarrow \Gamma(\cdot)$ is the standard Gamma function.
3. You may group the last paragraph in a remark.
4. p.3, line 2: give the two names for the references [MS21]. You use *et al* (with italic style) only when there are more than three authors.
5. p. 2, the equation for $\sigma_N(x)$, the word ”continuous and” is confusing. Should we just say that

$$\sigma_N(x) = \sigma \left(\left(1 \wedge \frac{N}{|x|} \right) x \right).$$

Response - I do not believe it is possible to write σ_N explicitly in higher dimensions. It may be more clear to just give σ_N is one dimension and then mention that it is defined similarly in higher dimensions as this is what is done in Millet and Sanz-Solé’s paper.

Le: The expression should work because when $N > |x|$, then

$$\left(1 \wedge \frac{N}{|x|} \right) x = x,$$

so that $\sigma_N(x) = \sigma(x)$. But when $N \leq |x|$, then

$$\left(1 \wedge \frac{N}{|x|} \right) x = N \frac{x}{|x|},$$

where $\frac{x}{|x|}$ is the normal vector and N is the radius. This is exactly the generalization from $d = 1$ to higher dimensional case $d \geq 2$. What do you think?

6. Last paragraph of Section 1, ”Itô” should be properly typed; I would say ”the $L^2(\Omega)$ existence” with ”the”.
7. Eq. (2.7), I would have $k > 0$ and write the second part as

$$\sup_{t \geq 0} \int_0^t s e^{-as} ds = \int_0^\infty s e^{-as} ds = \frac{1}{a^2} \quad \text{for all } a > 0$$

8. Regarding Proposition 2.4:

- (a) Did you forgot subscript 0 for J in the beginning of the proof? There also misses a period at the end of this centered equation. Check all thesis about such typos.
- (b) You may put the last two centered equation as one equation on p. 4.
Response - It will not fit

- (c) You may remove [MS21, Proposition 3.2] since you are going to prove this one. But you may mention the relation of this proposition with Proposition 3.2 of [MS21].

Response - I will remove these at the very end. I want to keep it for now as it allows me to quickly go back and compare the results.

- (d) First centered equation in the proof, you forgot the period after $\beta \in (1, 2)$
- (e) Eq. (2.10), $\sup_{t \geq 0} \sup_{x \in \mathbb{R}}$ should be $\sup_{t \geq 0, x \in \mathbb{R}^d}$.

- (f) It is cumbersome to put the two cases formula in other equations. Let's define the right-hand side of (2.10) by C_β and then simplify your presentation.

Response - I just realized that I have already defined that quantity as $\mathcal{T}_0$ in the statement of the proposition. I will use $\mathcal{T}_0$ to replace the 'cases' throughout the proof.

- 9. Proposition 2.7, put two conditions on a and p in one line (there is an unnecessary period).

- 10. Regarding the proof of Theorem 1.1:

- (a) the Picard iteration should be written with 1 replaced by $J_0(t, x)$ since you have equality here. Moreover, the upper bound of $J_0(t, x)$ is function of t in general even though we can make it uniform in x . But since we do not consider asymptotics in t , for fixed $t \in [0, T]$, we can say that $|J_0(t, x)| \leq C_T$ so that when we carry out the upper bound moment estimates, there won't be any issues.

- (b) The last sentence: This completes the proof of Theorem 1.1.

- (c) The rate $\frac{\ln(2N)^{p\delta/2}}{N^p}$ makes one think the result is far from being sharp. we are using polynomial to combat against logarithm. This is unfair even this does prove our results. Can you check [MS21] and Dalang's original work to see if this is indeed the case? There have to be a lot room to improve such result.

Response - this is the same as [MS21] and equation (4.7) in Dalang's paper suggests that Dalang proves the same thing as well.

- (d) The two-line equation for $\mathbb{P}(\tau_N < T)$ can be made in one line; the next two-line equation looks too wide. Try to improve the presentation of the paper.
- (e) At beginning of Section 4, always find a proper place to introduce your convention on Fourier transform.

Response - Put it in the intro.

- (f) Assumption 4.1, Use period at each end of items and capitalize the initial letter.